

ANEO

Balancing of the Nordic Power System

Bjørn H. Bakken

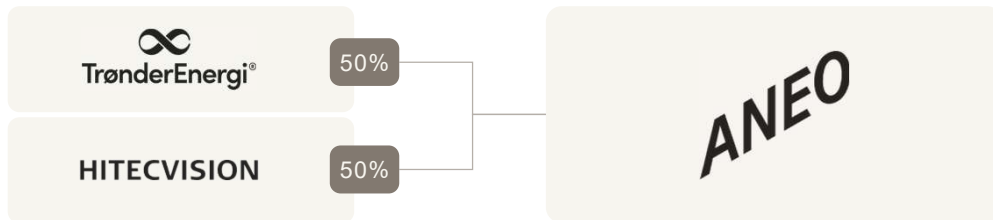
Head of Physical Trading, PhD

Aneo AS



About Aneo

ABOUT ANEO



Norway
Sweden
Finland



Asset
ownership
34
power plants
across the
Nordic region



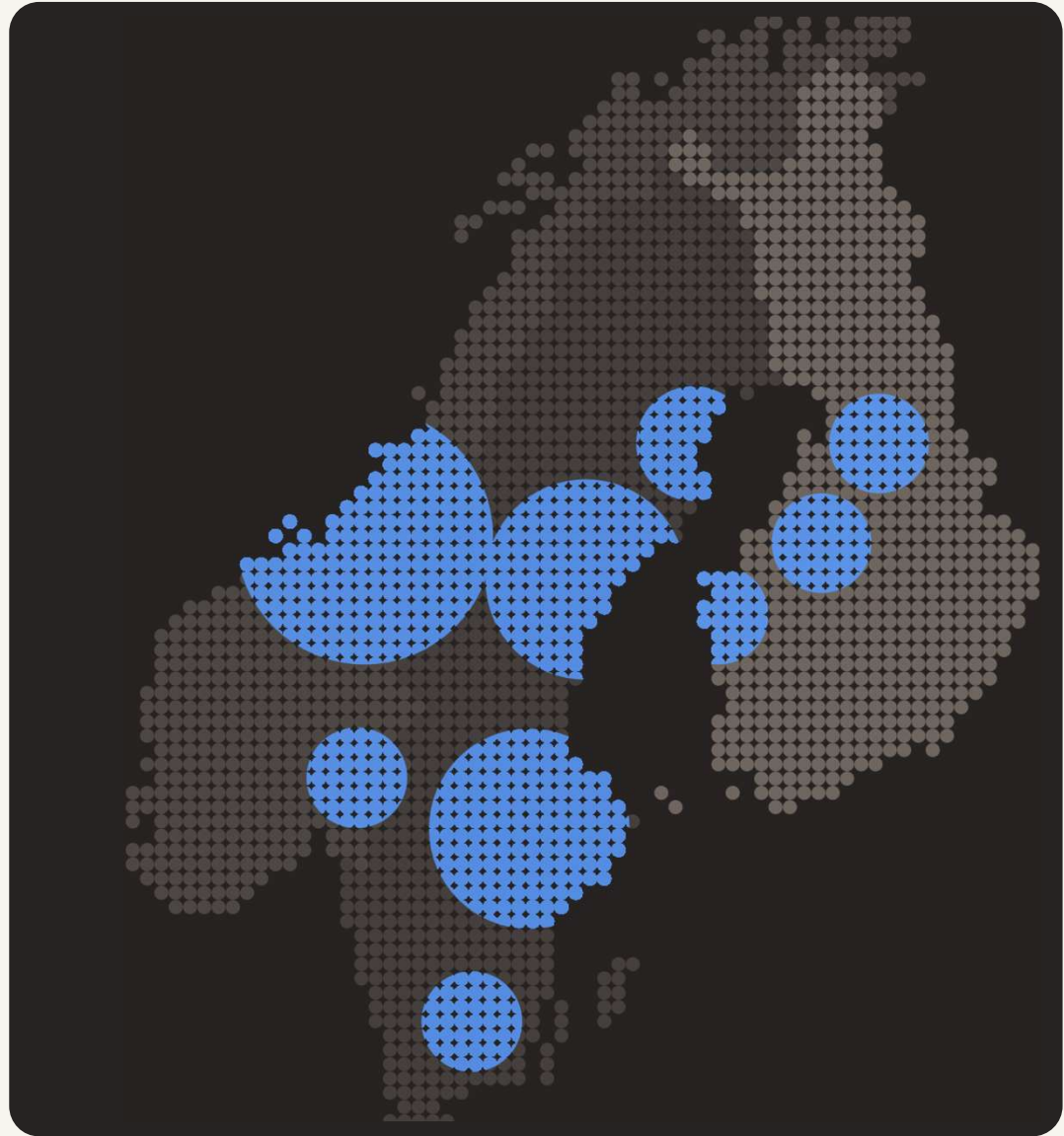
Employees
~200

Locations
16



Energy
Markets

3300 MW
11 TWh



Agenda

1. The sequence of power markets
2. Why is the frequency important?
3. Nordic Balancing Model (NBM): A new way of balancing
4. Consequences: Increased system costs
5. Value stacking: New commercial opportunities

Agenda

1. The sequence of power markets

2. Why is the frequency important?
3. Nordic Balancing Model (NBM): A new way of balancing
4. Consequences: Increased system costs
5. Value stacking: New commercial opportunities

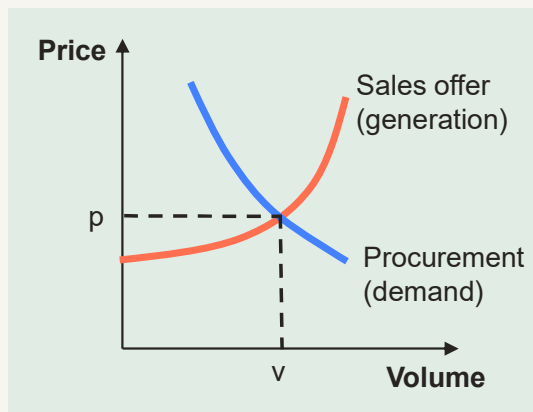
Stable operation of the power system is ensured by a sequence of markets



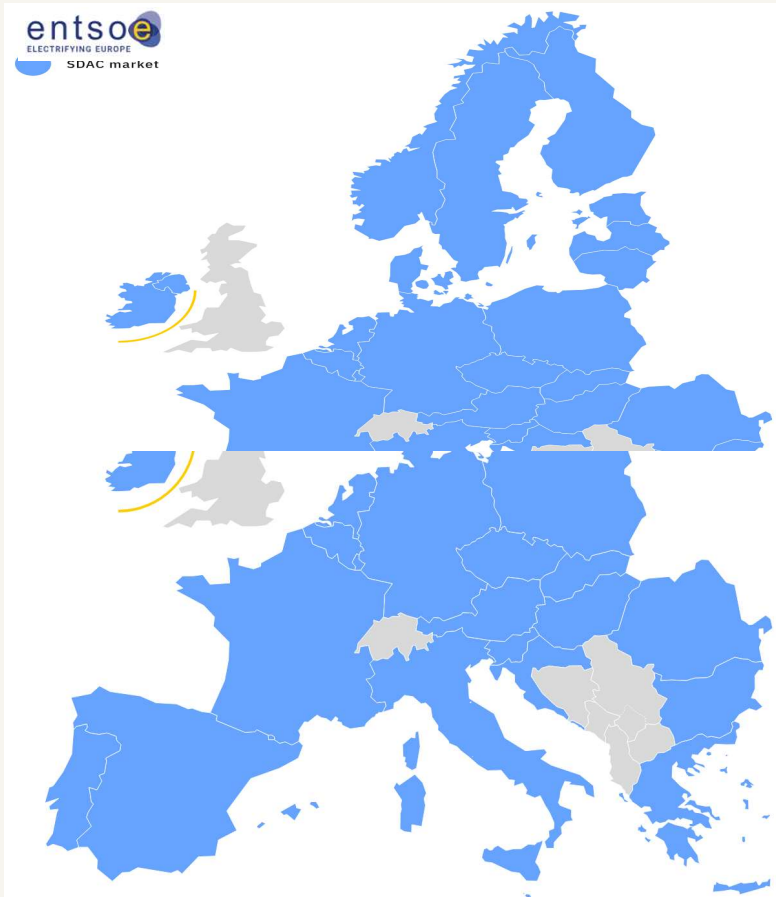
SDAC – Single Day-Ahead Coupling
IDC – Intraday Continuous
IDA – Intraday Auction

The Spot market establishes an initial energy balance

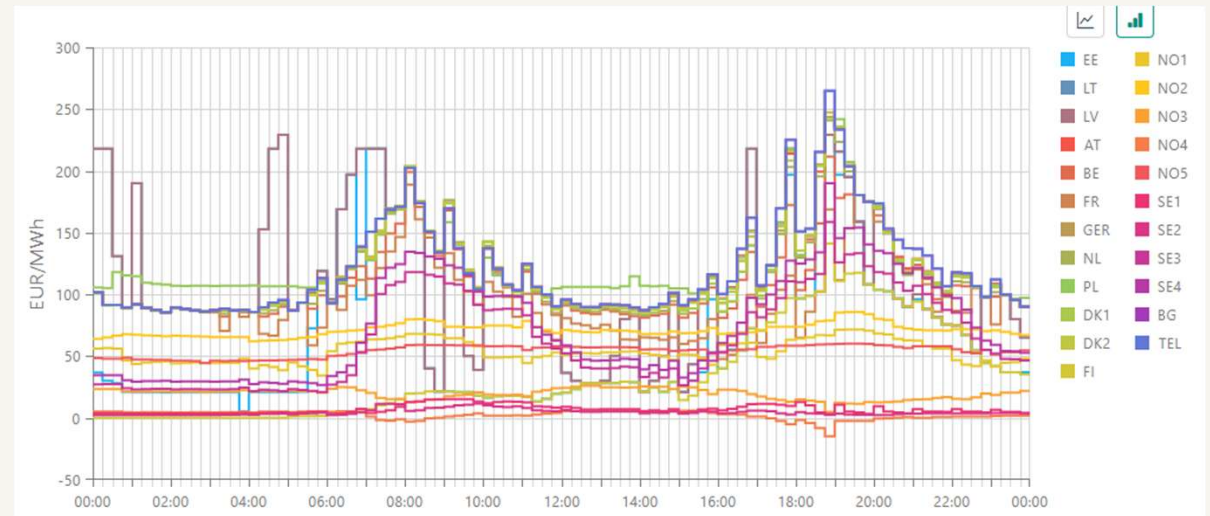
- The power system must always be in balance to maintain system stability. The sum of production, consumption and net import/export must be zero at any time.
- **The Spot market establishes a common price based on the balance between planned production and demand each 15 minutes.**
- The energy market ensures that total production resources are utilized in a cost-effective way since the cheapest production is traded first. In the longer term, the energy prices contribute to cost effective investments in production and consumption.



Most of Europe is connected in the **Single Day-Ahead Coupling** (SDAC)

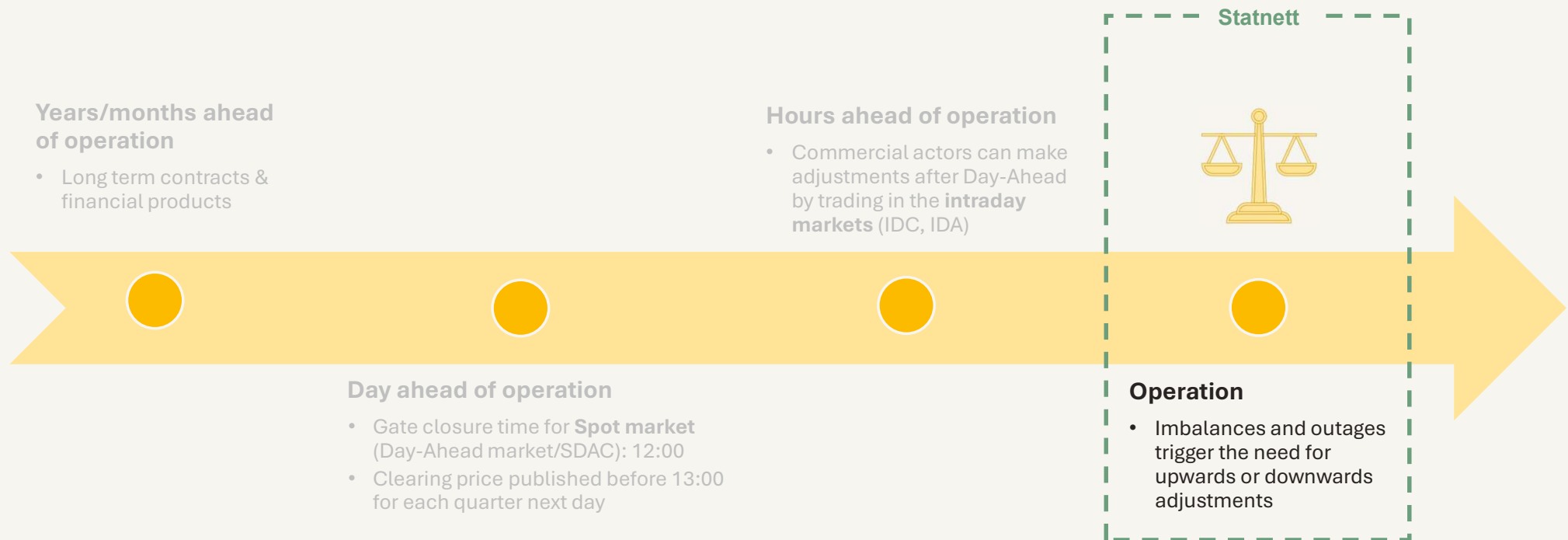


...where prices and exchange are calculated for each area for each quarter of the day



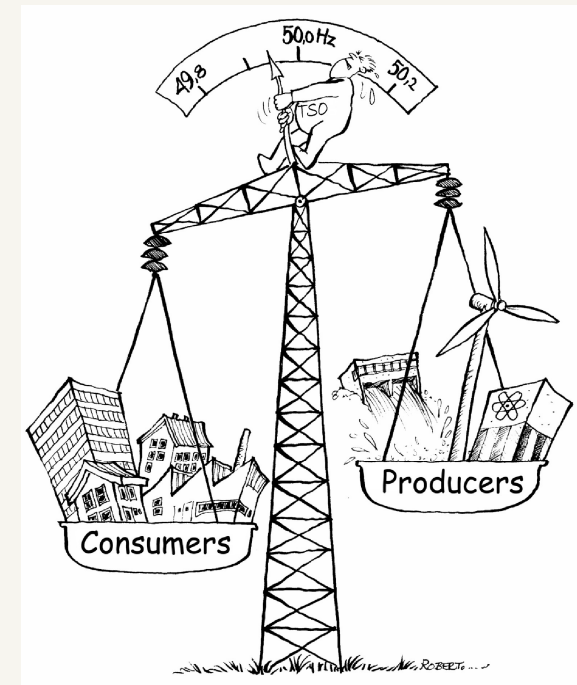
Nord Pool | Day-ahead prices

Stable operation of the power system is ensured by a sequence of markets

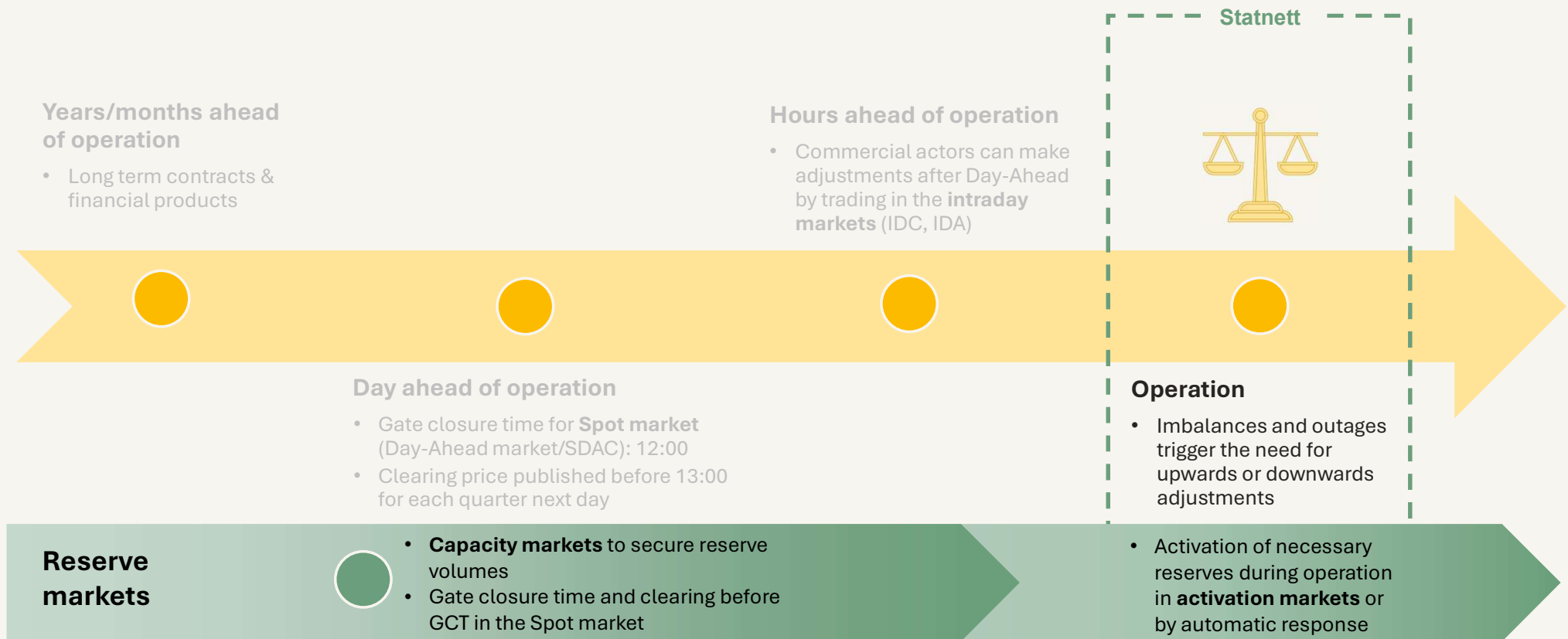


Imbalances between production and consumption are handled by the TSO

- The energy markets (Day-Ahead and Intraday) ensure that the power system is balanced **before delivery**. However, there will always be some deviations in actual production or consumption, and outages can occur.
 - **This causes imbalances**
- Imbalances are handled by the Norwegian Transmission System Operator (TSO) Statnett, which buys different products in **reserve markets**.
- The participants in the reserve markets are paid to adjust their consumption or production when needed. Some reserve markets require automatic solutions, fast responses and short duration, while others require more energy for a longer period of time.
 - Statnett secures sufficient reserves through **capacity markets**
 - The reserves are activated during the delivery period automatically or through specific **activation markets**



Stable operation of the power system is ensured by a sequence of markets



Agenda

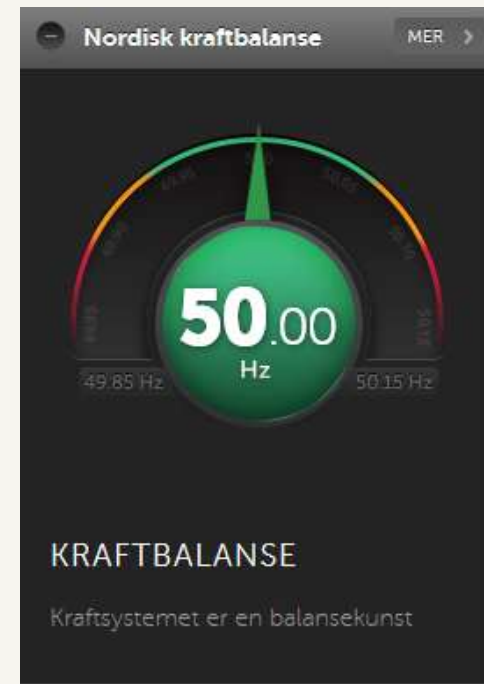
1. The sequence of power markets
- 2. Why is the frequency important?**
3. Nordic Balancing Model (NBM): A new way of balancing
4. Consequences: Increased system costs
5. Value stacking: New commercial opportunities

Why is the frequency important?

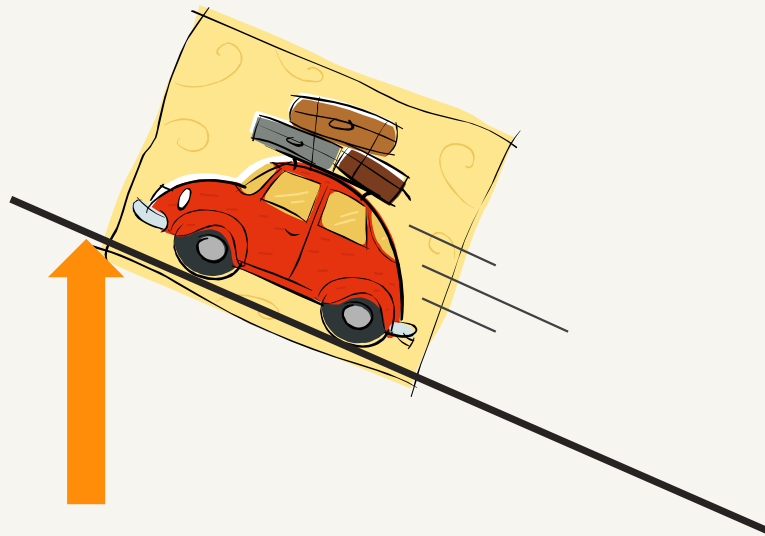
- **Frequency is a measure of the balance of the system**
- Imbalances cause unplanned changes in power flows in the system
- In case of imbalance, the frequency changes
 - More consumption than production leads to a decline in the frequency
 - More production than consumption leads to an increase in frequency
- If the frequency drops too much, generators can switch off automatically. This leads to a further decline in frequency and further disconnections. This spiral creates a risk of system collapse.
- Statnett uses reserves to keep the frequency at nominal value – **50 Hertz (Hz)**



Flat road => Constant throttle yields constant speed...



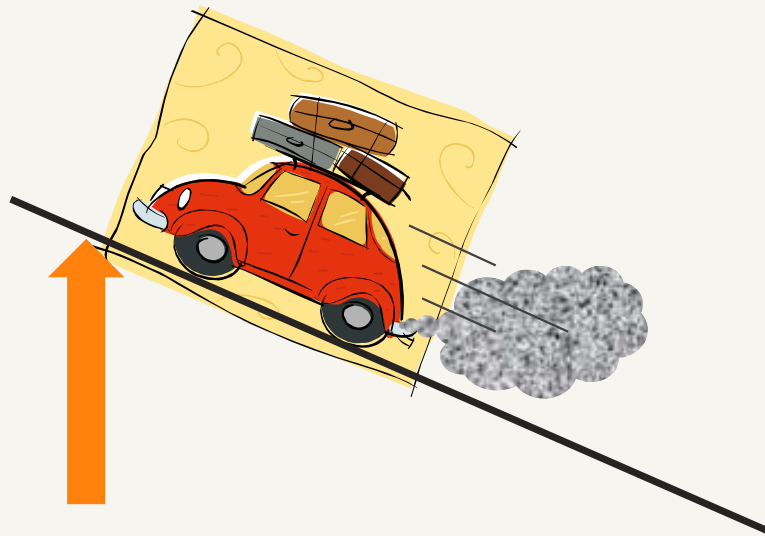
Uphill => The car will slow down



Demand
increases



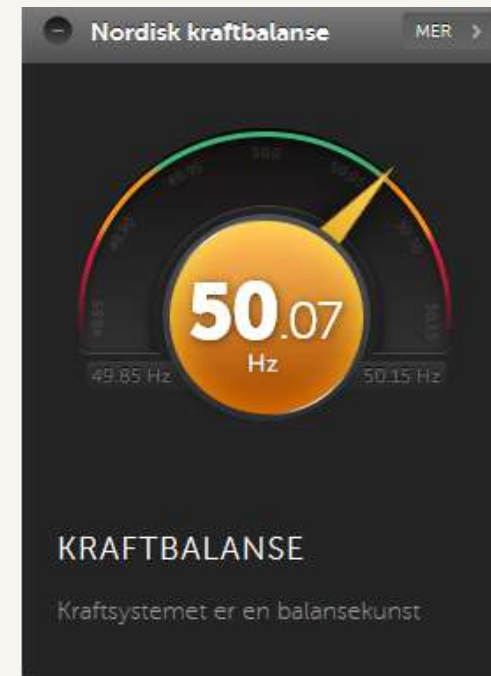
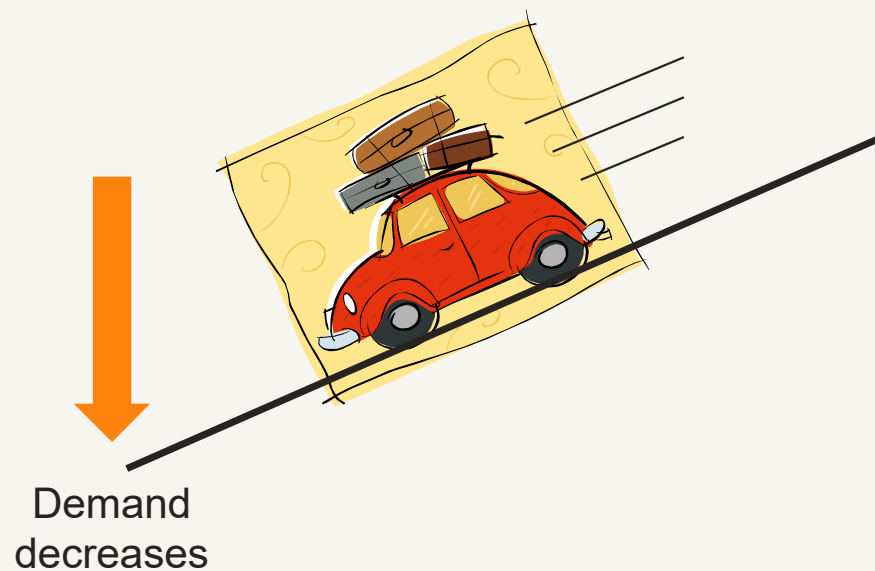
Uphill => The car will slow down
- *unless you accelerate*



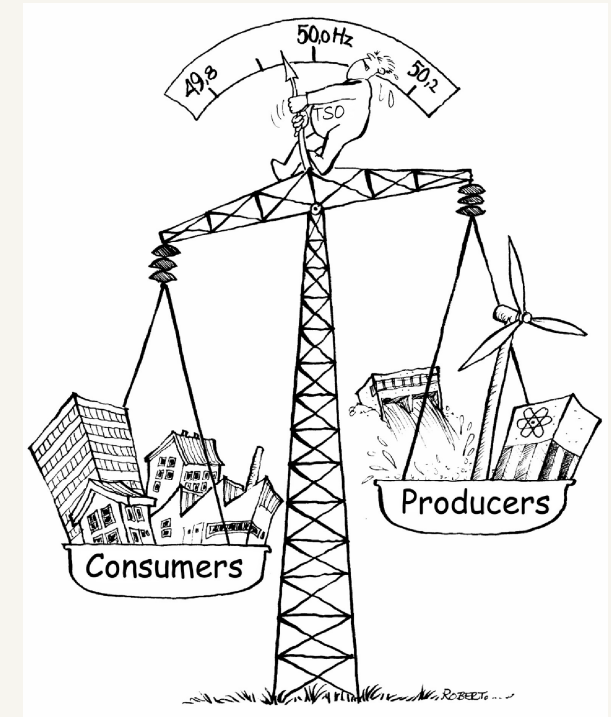
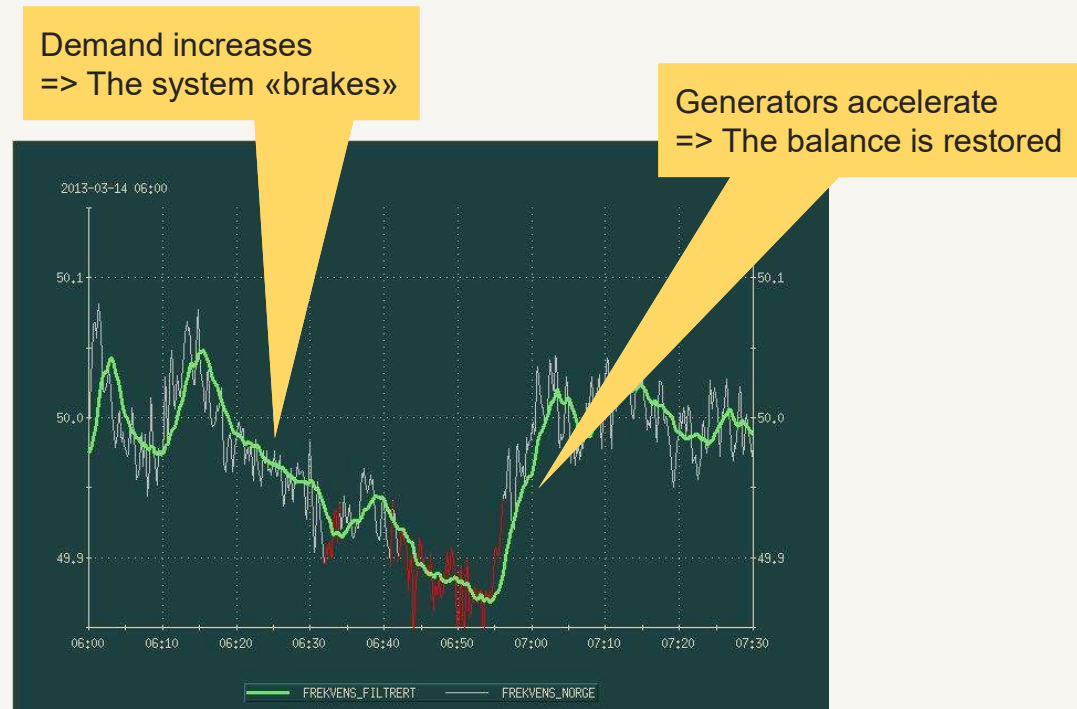
Demand
increases



In the same way, the car speeds up downhill
unless you brake



The frequency is a measure of the (im)balance between generation and demand



Statnett buys several types of reserves to maintain frequency

- with different types of suppliers



- **Fast Frequency Reserves FFR**

- 0,7-1,3 seconds response; 5-30 seconds duration
- Local frequency triggers the response
- *Capacity market and compensation for activation*



- **Frequency Containment Reserves FCR** (primary reserves)

- 30 – 120 seconds response; minimum 15 minutes duration
- Continuous response in proportion to local frequency
- *Capacity market and compensation for activated volume*



- **Automatic Frequency Restoration Reserves aFRR** (secondary reserves)

- Full response in 5 minutes; duration as long as the bid period lasts
- Continuous control by SCADA-connection between Statnett and production unit
- *Capacity market and compensation for activated volume (activation market upcoming)*

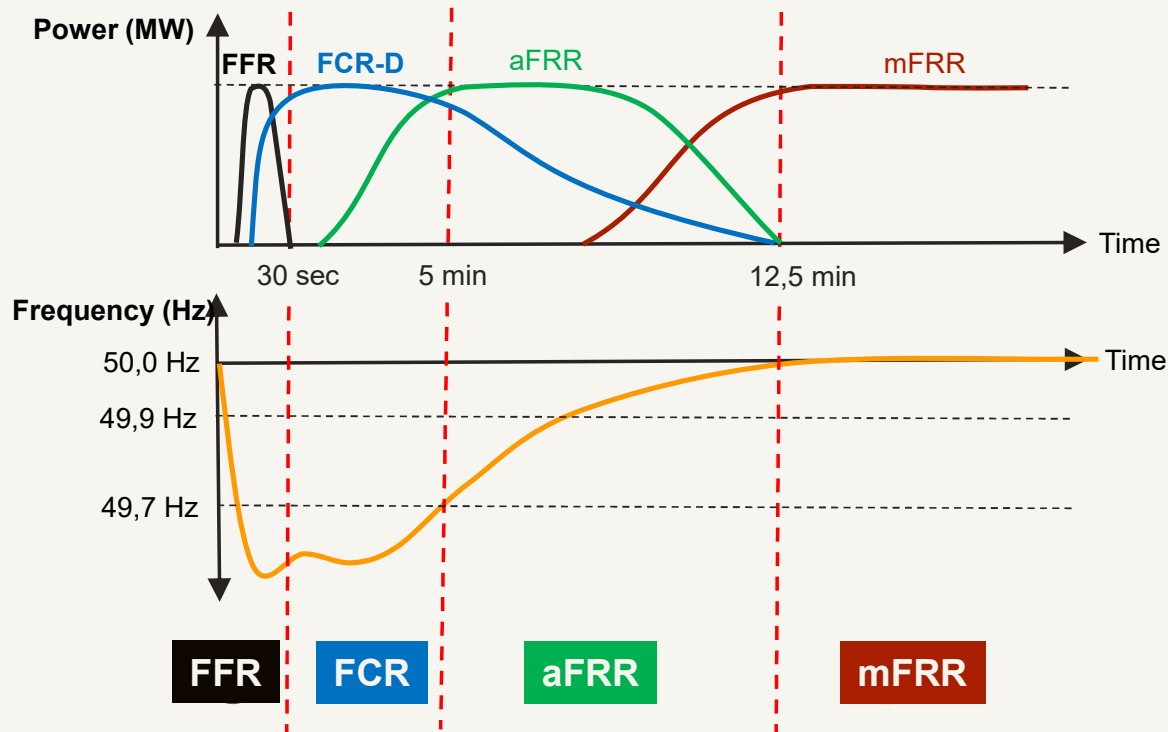


- **Manual Frequency Restoration Reserves mFRR** (tertiary reserves)

- Full response in 12,5 minutes; duration as long as the bid period lasts
- Electronically activated every 15 minutes
- *Capacity market (mFRR CM) and activation market (mFRR EAM)*

Which reserves are used when?

«Classic» illustration of reserve response after an outage



FFR Slows down the frequency drop

FCR Contains (stops) the frequency drop and stabilizes the frequency at a new level

aFRR Restores frequency back towards the nominal frequency (50,0 Hz)

mFRR Restores frequency, releases aFRR and maintains balance until new balance is reached in the energy market

Agenda

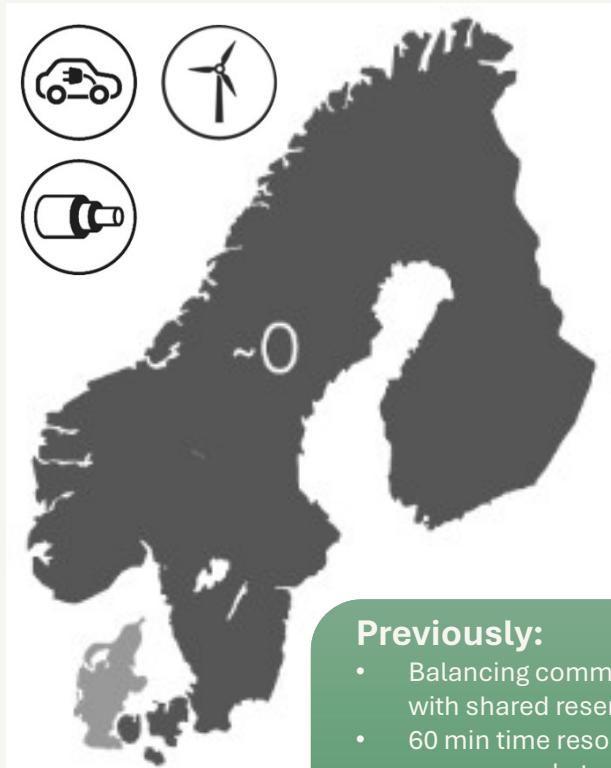
However...

1. The sequence of power markets
2. Why is the frequency important?
- 3. Nordic Balancing Model (NBM): A new way of balancing**
4. Consequences: Increased system costs
5. Value stacking: New commercial opportunities

Nordic Balancing Model

A new way of balancing the Nordic power system

NBM
Nordic Balancing
Model



Previously:

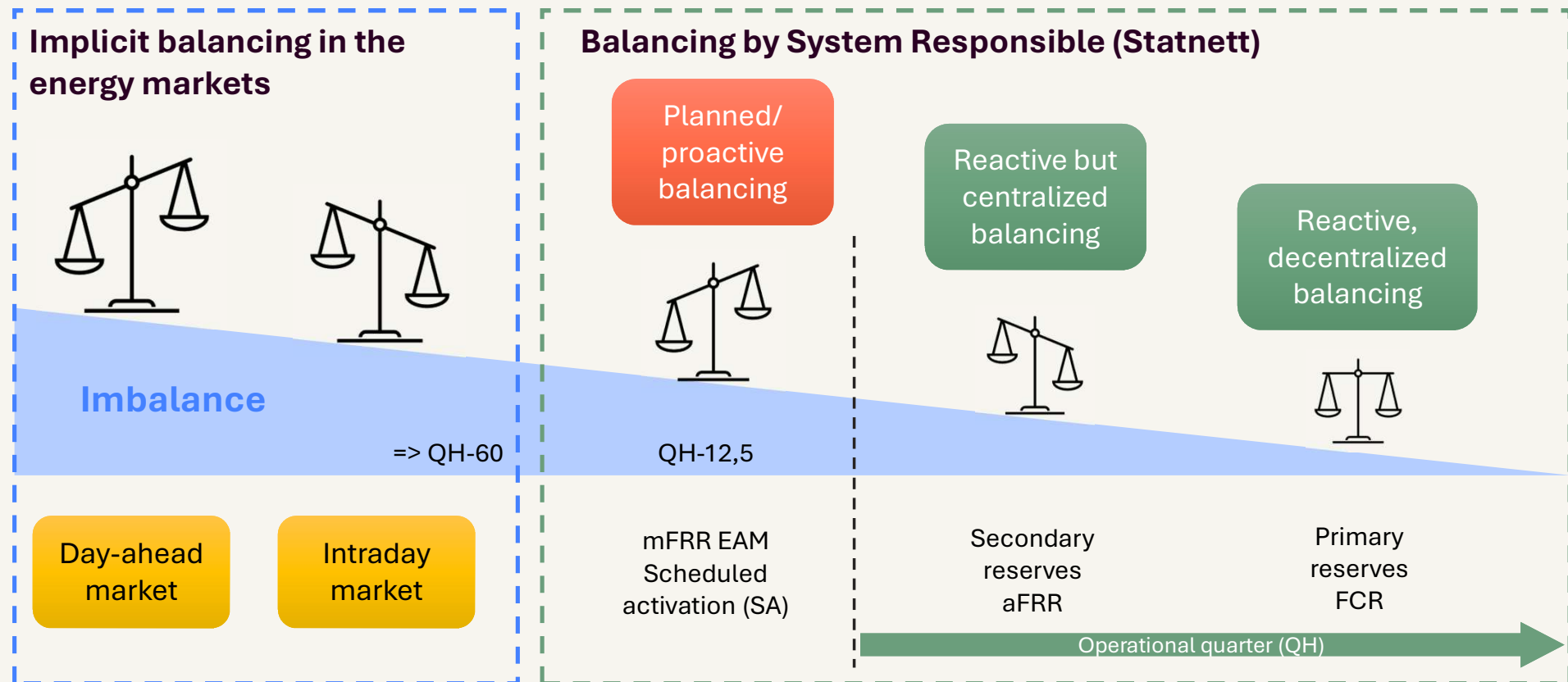
- Balancing common Nordic frequency with shared reserves
- 60 min time resolution in energy and reserve markets
- Manual interventions from operators
- Congestion and balancing handled simultaneously



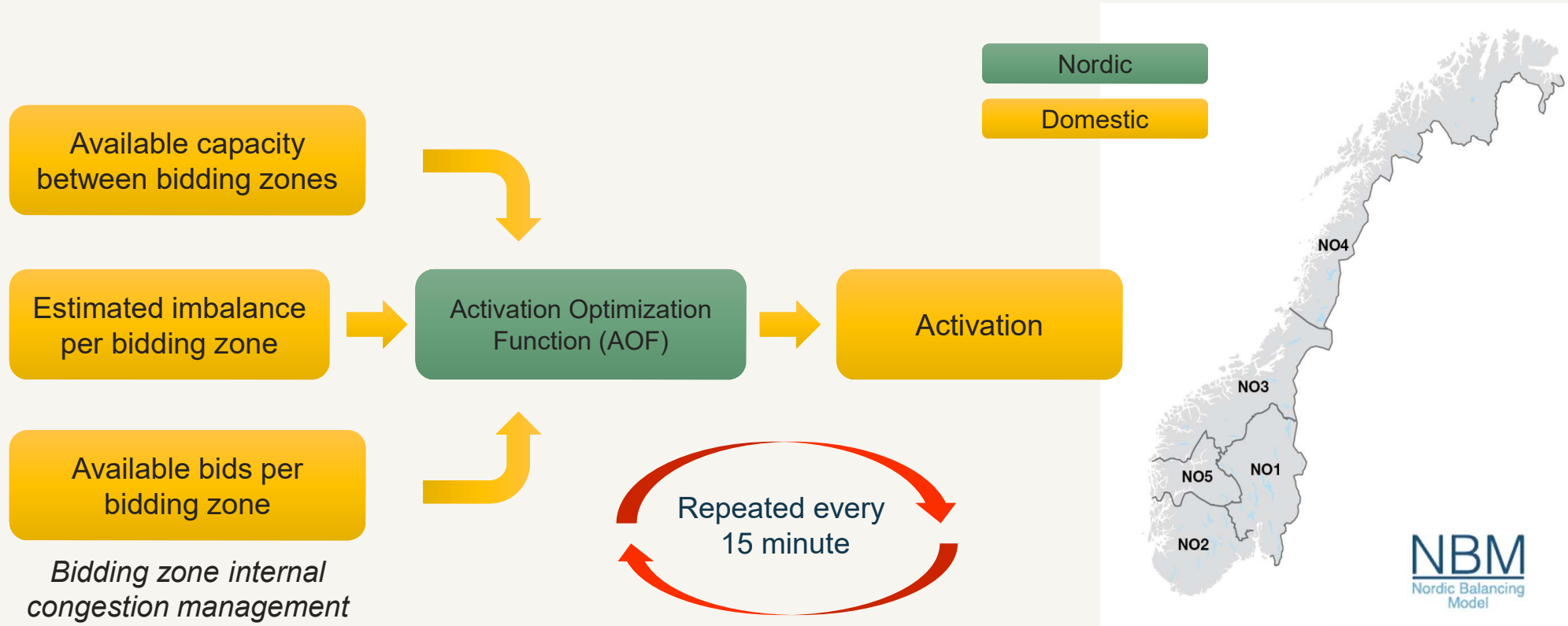
Today:

- Each bidding zone balanced separately
- 15 min time resolution in markets
- Automatic solutions to support operators
- Consistent with European balancing platforms

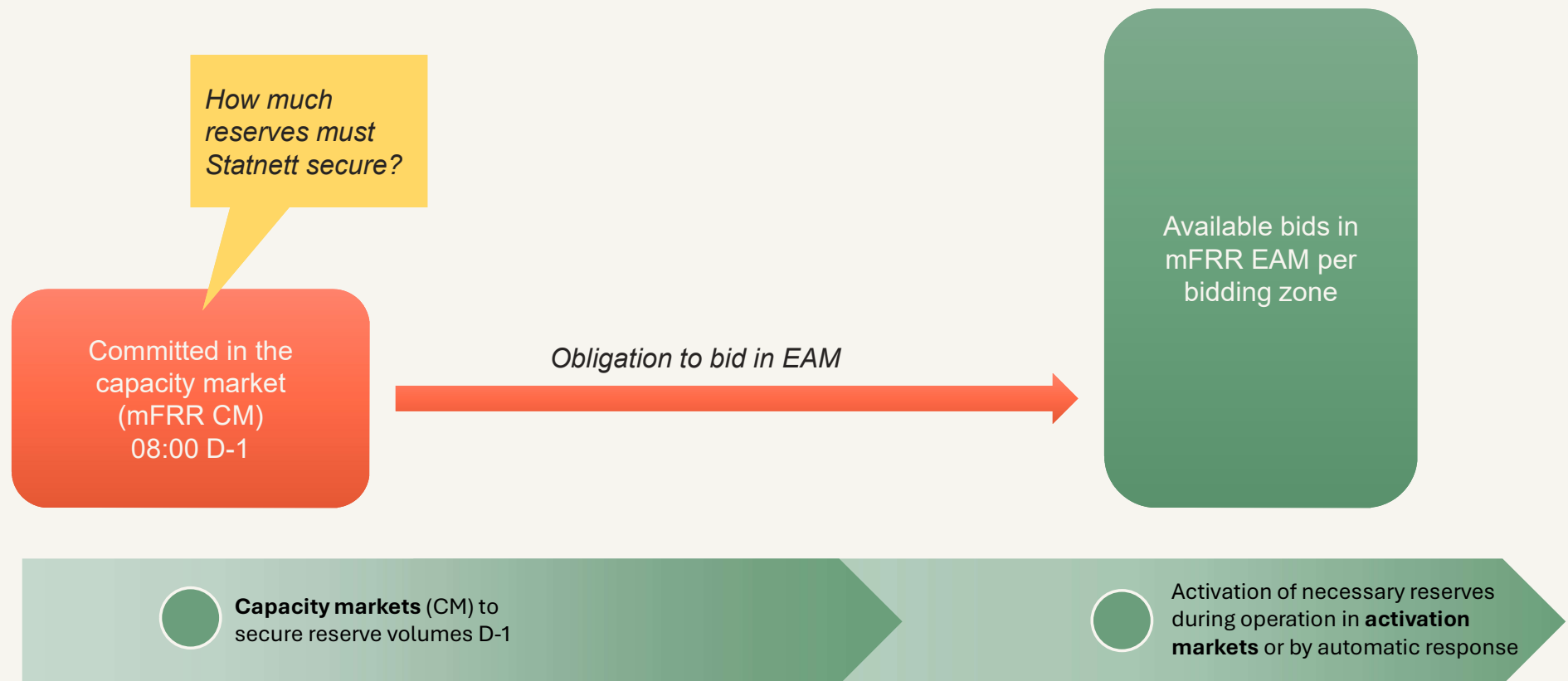
We have moved from *reactive* to *proactive* balancing



The automatic balancing process (mFRR Energy Activation Market EAM)



Capacity secured in CM has an obligation to bid into EAM



How much reserves must Statnett secure?

Dynamic dimensioning of mFRR capacity needs

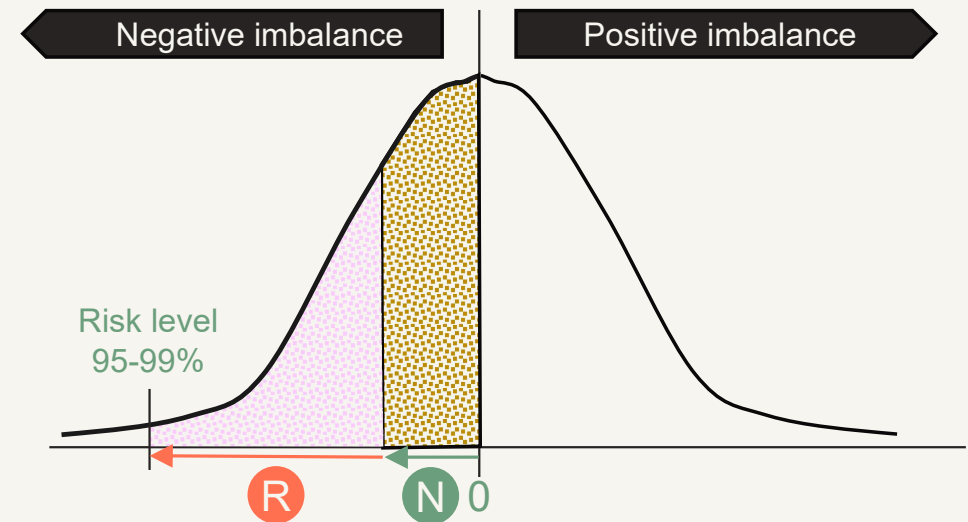
Definitions

N **Normal Imbalances** occur when producers or consumers deviate from their plans

R **Reference Incident** is the largest single outage of a grid component, HVDC connection, generator or load that may occur in a bidding zone

S **System regulations** are activations of reserves by Statnett to avoid overloading components inside the bidding zone

Probability distribution of imbalances



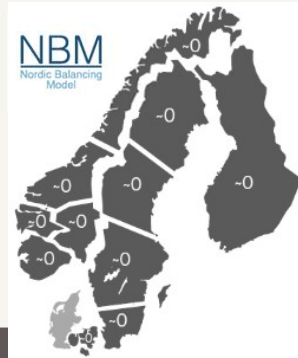
How much reserves must Statnett secure?



Historical model



Balancing common Nordic frequency
with shared reserves
Only upwards regulation secured



Automatic
balancing
(mFRR EAM)
March '25



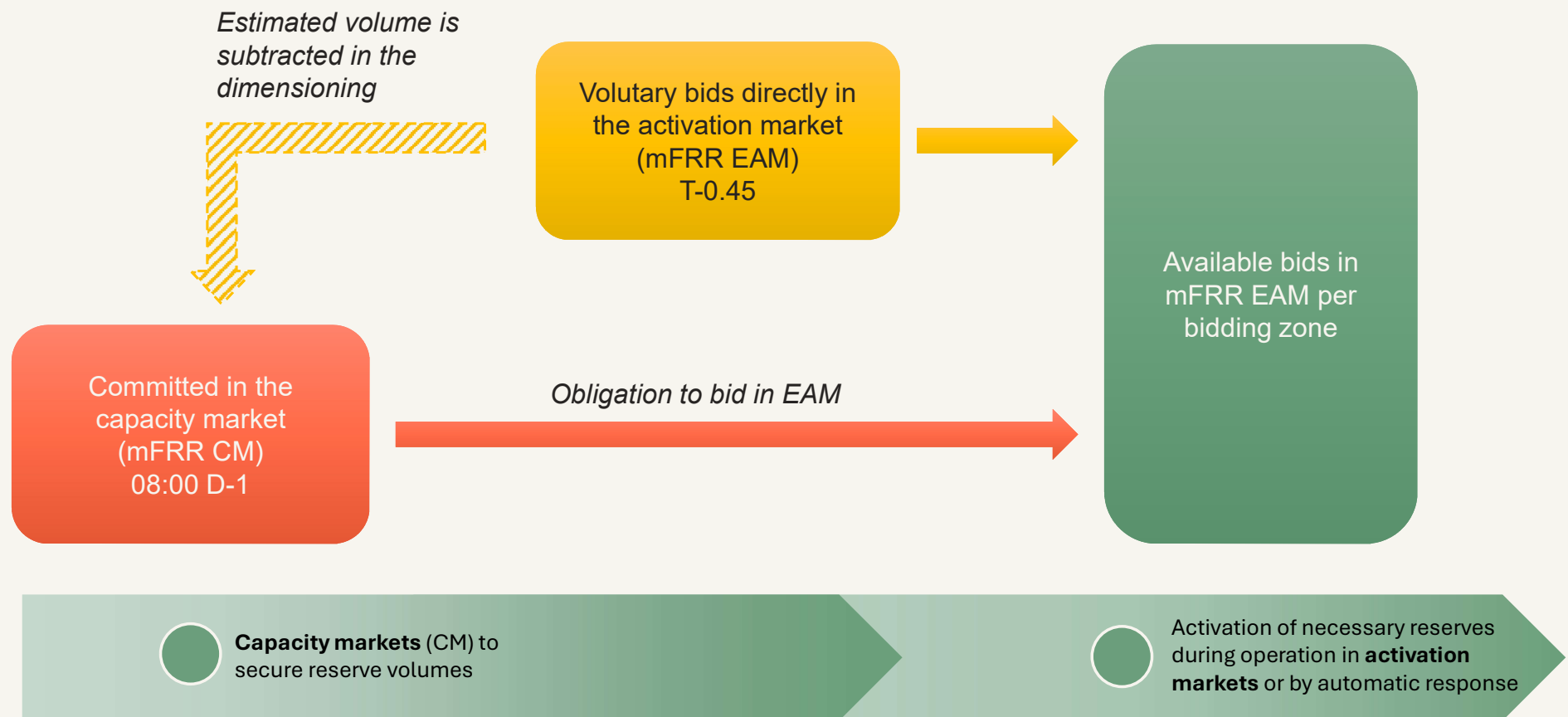
Balancing per bidding zone
Imbalances and outages in both
directions must be handled
90-95% of the time

European balancing
markets
(MARI/PICASSO)
2027/2028

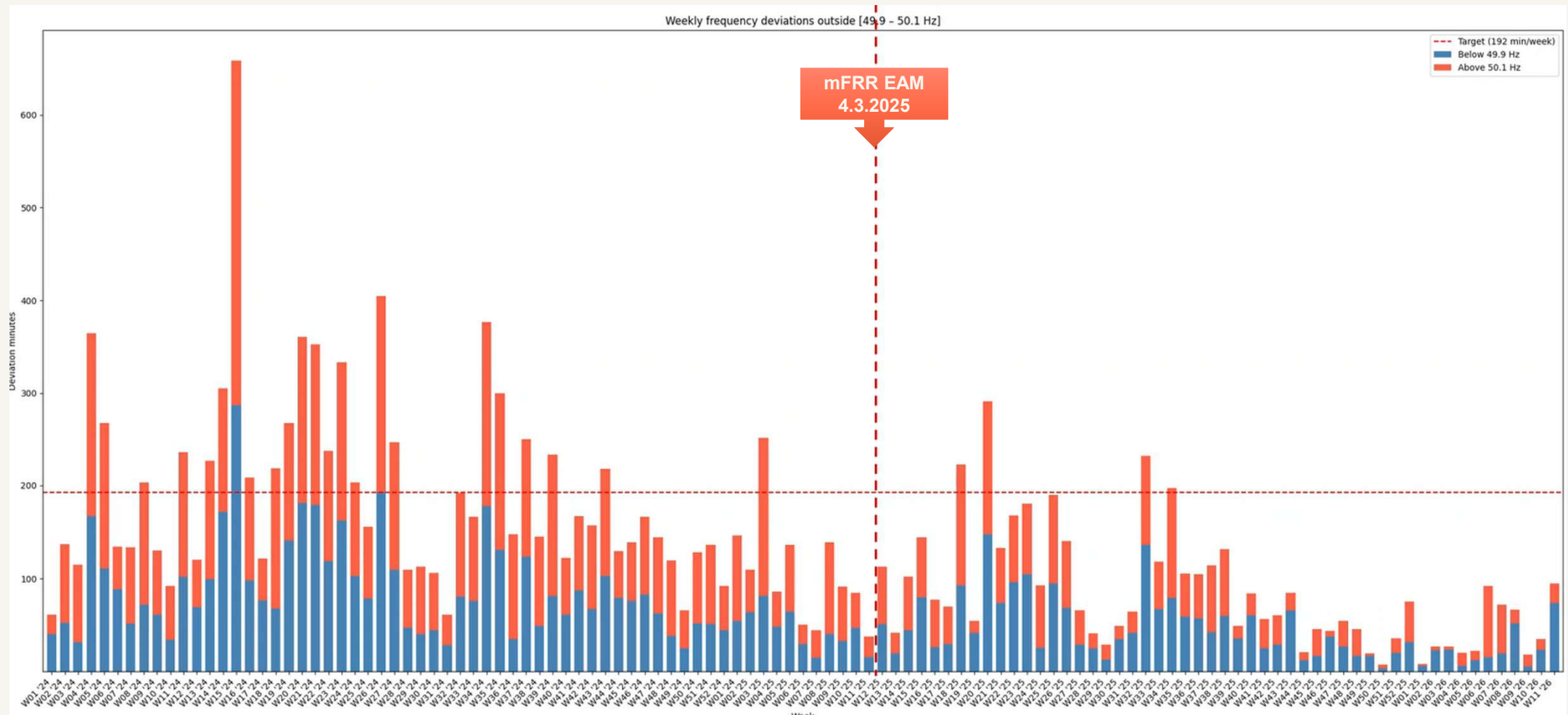


Balancing per bidding zone
Imbalances and outages in both
directions must be handled
99% of the time

Available bids in mFRR EAM are the sum of capacity secured in CM plus “voluntary bids” submitted directly in EAM



Frequency quality has improved after mFRR EAM go-live - *but at what cost?*

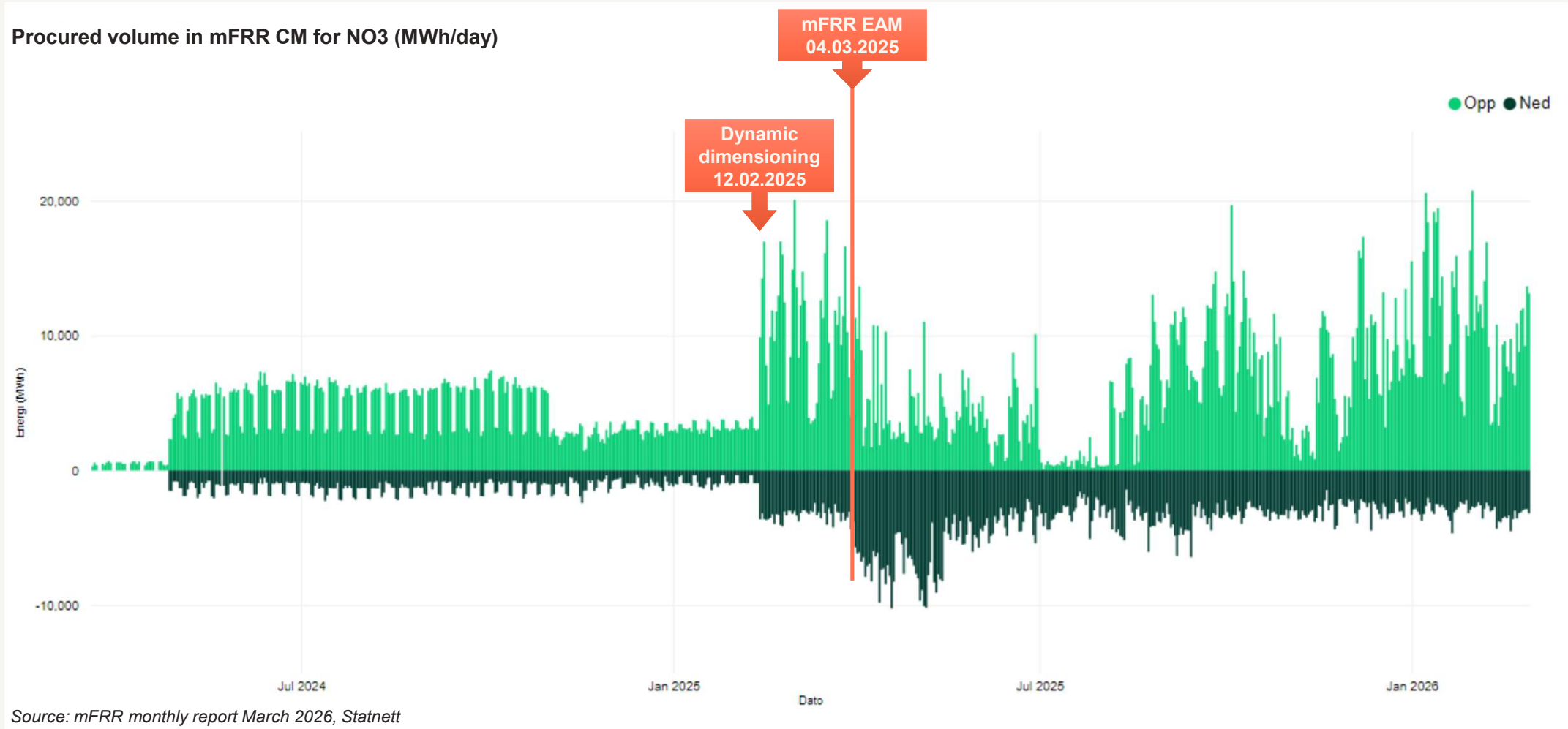


28 Source: Presentation of mFRR monthly report March 2026, Statnett

Agenda

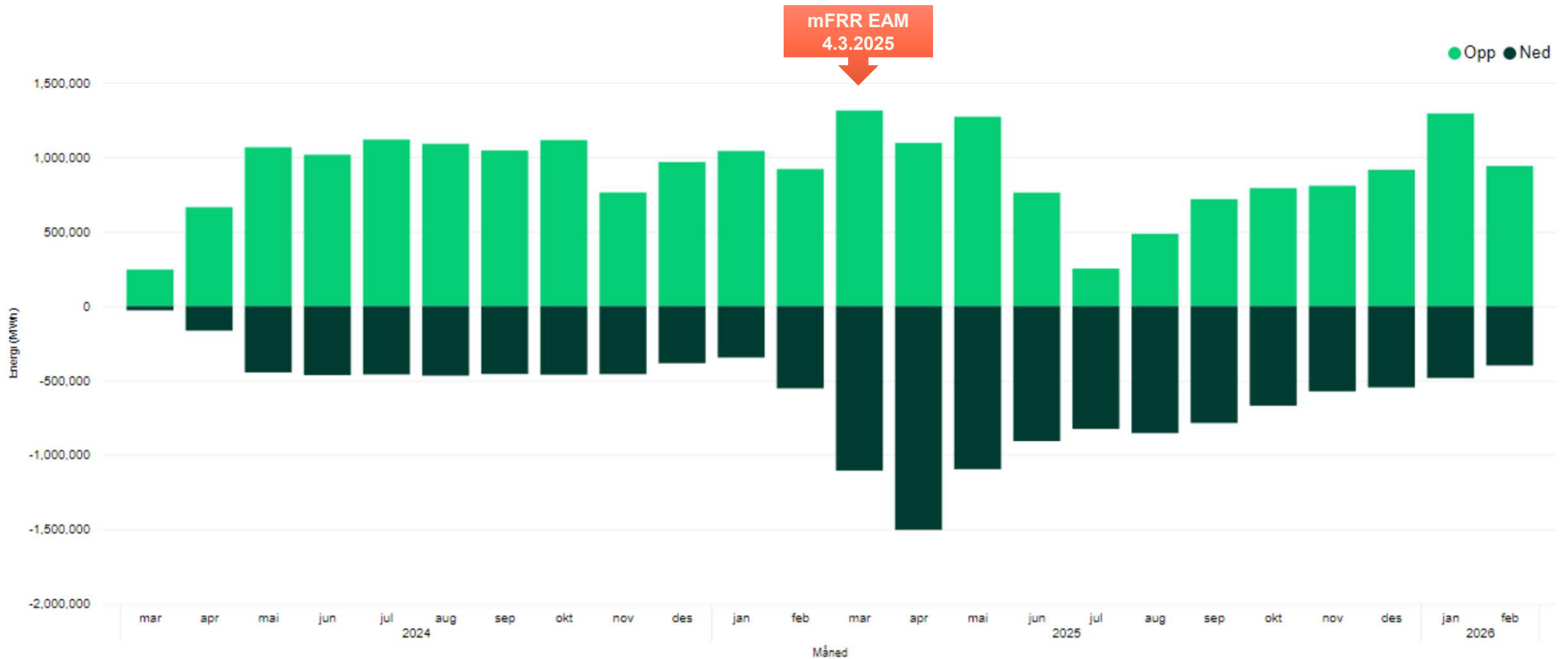
1. The sequence of power markets
2. Why is the frequency important?
3. Nordic Balancing Model (NBM): A new way of balancing
- 4. Consequences: Increased system costs**
5. Value stacking: New commercial opportunities

Procurement of capacity changed dramatically after mFRR EAM go-live in March 2025



Changes are less visible on a national scale (MWh/month)

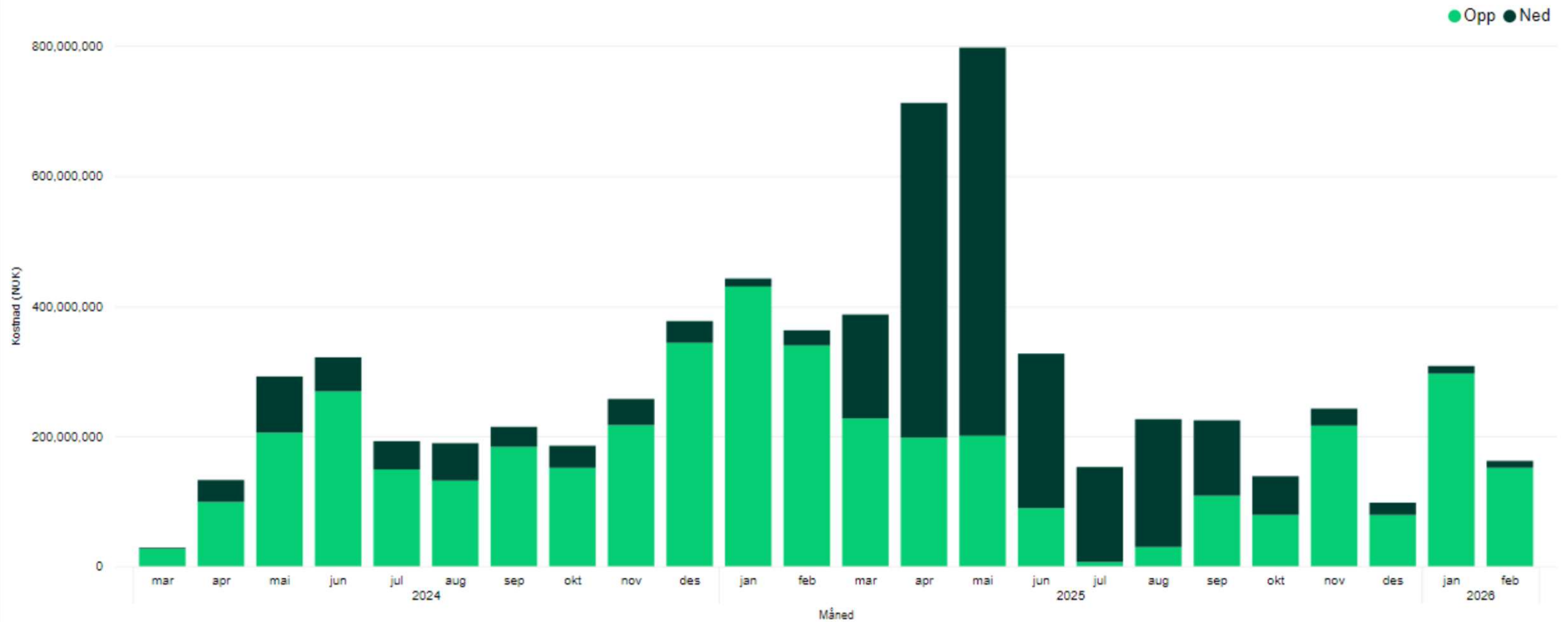
Procured volume in mFRR CM for all Norwegian bidding zones (MWh/month)



Source: mFRR monthly report March 2026, Statnett

We have seen large volatility in system operation costs

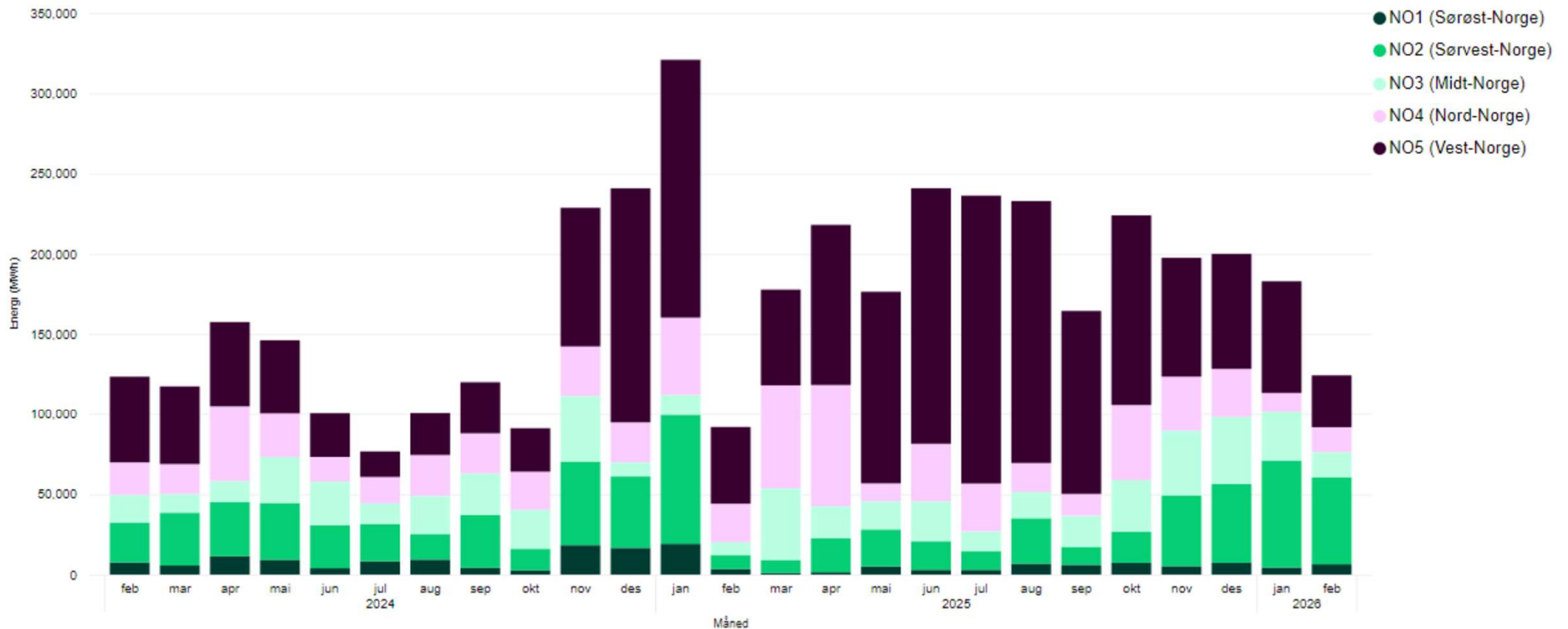
Cost of mFRR CM for all Norwegian bidding zones (NOK/month)



Source: mFRR monthly report March 2026, Statnett

Activated volumes for upwards regulation have increased with mFRR EAM

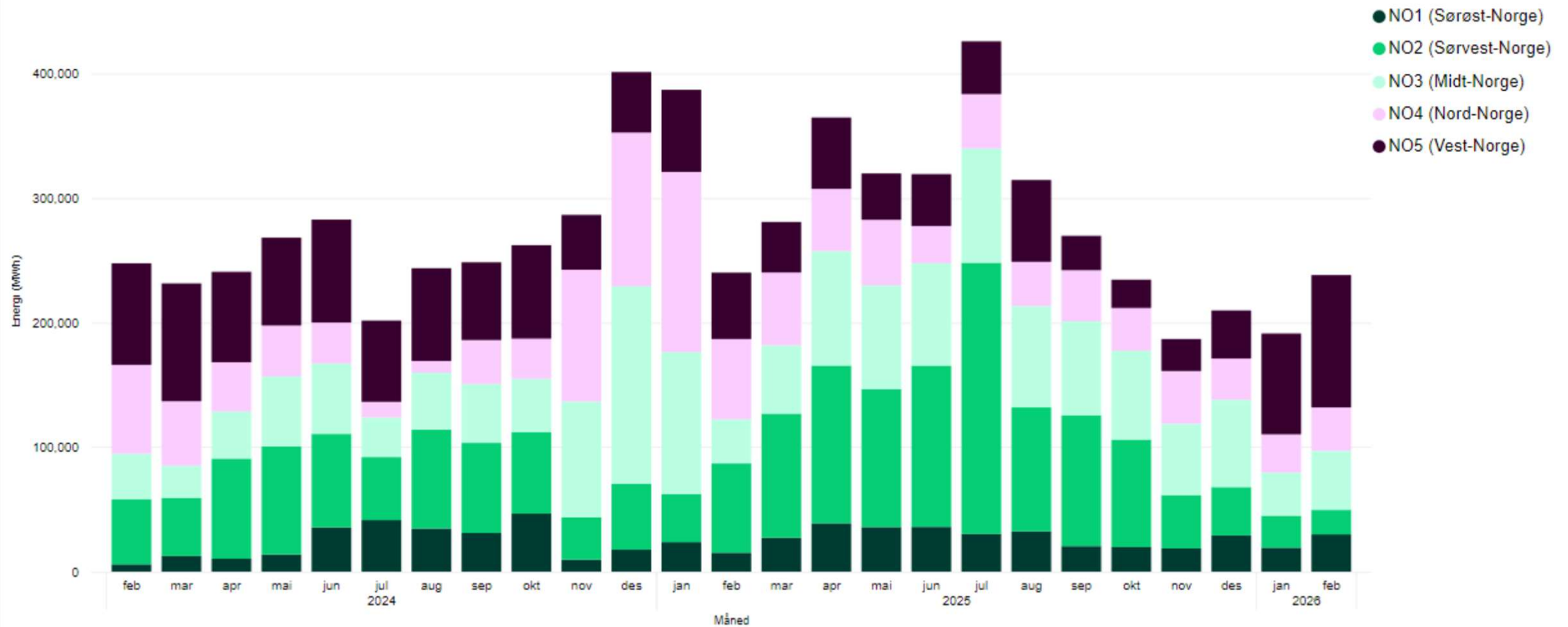
Activated volumes in mFRR EAM up for all Norwegian bidding zones (MWh/month)



Source: mFRR monthly report March 2026, Statnett

Activated volumes for downwards regulation not so much

Activated volumes in mFRR EAM down for all Norwegian bidding zones (MWh/month)



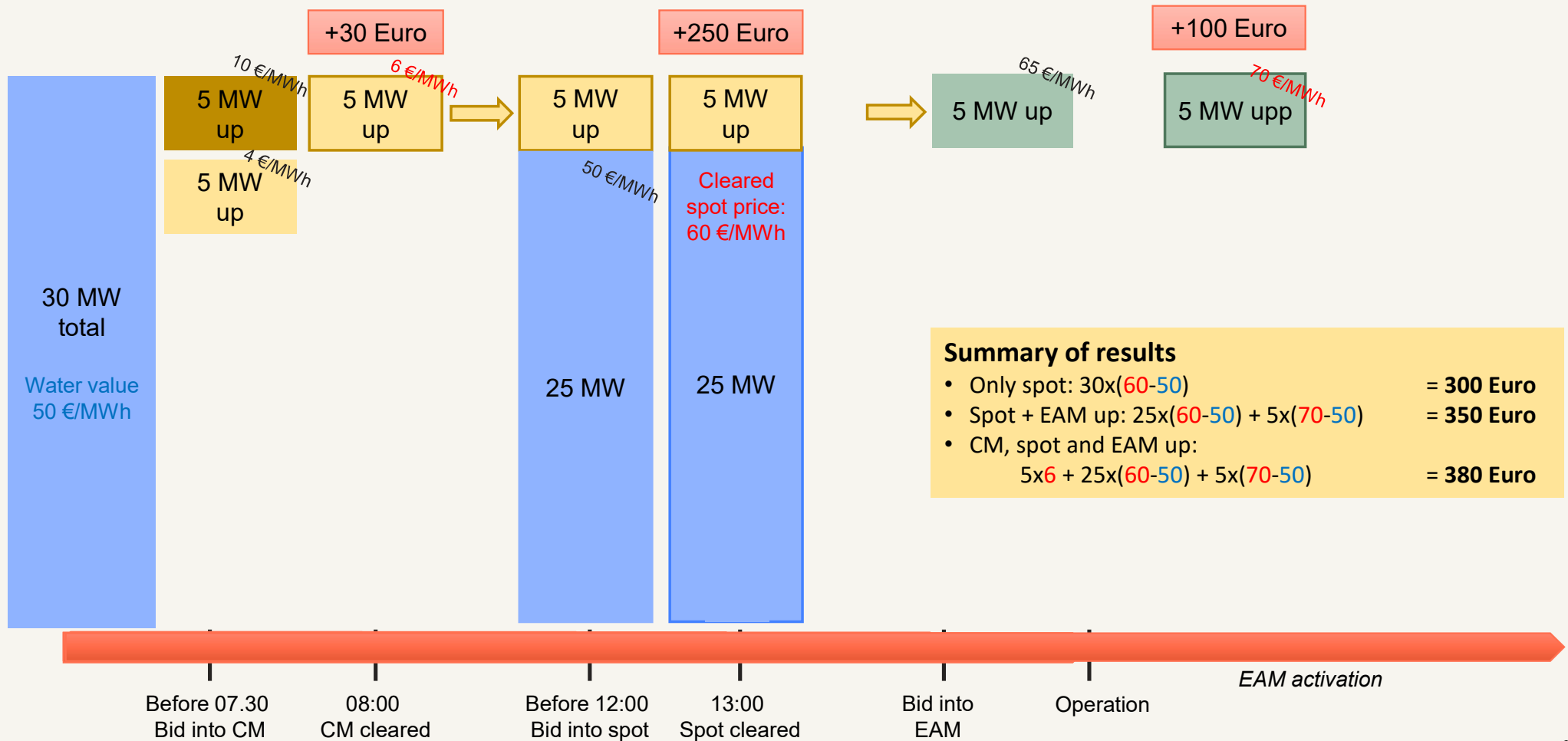
Source: mFRR monthly report March 2026, Statnett

Agenda

1. The sequence of power markets
2. Why is the frequency important?
3. Nordic Balancing Model (NBM): A new way of balancing
4. Consequences: Increased system costs
- 5. Value stacking: New commercial opportunities**

Reserve markets offer additional income to generators

Example of **value stacking** with hydropower in multiple markets



Further reading

- Information about the reserve markets (in Norwegian)
 - <https://www.statnett.no/for-aktorer-i-kraftbransjen/systemansvaret/kraftmarkedet/reservemarkeder/>
- Interactive reserve market calculator (in Norwegian)
 - <https://www.statnett.no/for-aktorer-i-kraftbransjen/systemansvaret/kraftmarkedet/reservemarkeder/reservemarkedskalkulator/>
- Nordic Balancing Model
 - <https://nordicbalancingmodel.net/>
- Nordic report "*Wind and solar in the Nordic reserve markets*"
 - <https://www.statnett.no/globalassets/for-aktorer-i-kraftsystemet/marked/reservemarkeder/nordic-wind-and-solar-publication-.pdf>
- Nordic report "*Batteries in the Nordic reserve markets*", April 2025
 - <https://www.statnett.no/globalassets/for-aktorer-i-kraftsystemet/marked/reservemarkeder/nordic-tsos---batteries-in-the-nordic-reserve-markets-april-2025.pdf>
- Thematic report "*Fleksibilitet som kilde til verdiskaping og forretningsutvikling*" (in Norwegian)
 - <https://www.statnett.no/globalassets/for-aktorer-i-kraftsystemet/systemansvaret/reservemarkeder/fleksibilitet-som-kilde-til-verdiskaping-og-forretningsutvikling.pdf>

ANEO

Balancing of the Nordic Power System

Bjørn H. Bakken
Head of Physical Trading, PhD
Aneo AS

Thank you for your attention!

